



Fun with dry ice

Dry ice is far from being dry when it comes to having fun with it
<http://dgit.in/16C8dxp>



Woopsie!

Apple recently pushed an update for Find my iPhone limited to developer accounts only
<http://dgit.in/177W5D8>

Workshop

for an 8 channel read-out, they speculated the presence of 2 amplifier circuits and designed the code that would program them to operate at two independent ISOs.

What Dual ISO is Actually Doing

When the mode is activated, it is essentially capturing two ISO frames at the same time. However, the image is quite garbled, so the light and dark frames must be separated and then recombined with usable information from both frames to yield one proper image. The downside for photo of this process is that the image yielded is technically a half resolution file (the image will still read a full 21 megapixels, but if you zoom in at 100%, you notice scan lines). For web-work, it shouldn't matter, but if you're looking to make billboard sized prints, we recommend using the bracketing approach to capture maximum dynamic range. As for video, it should be obvious that since it's a frame-by-frame alternation of ISOs, if you get motion in your footage, it will look unnatural. However if you want to shoot a video for interiors, go right ahead and exploit this technique.

Okay, Okay, I get it! How Do I Get it on my Camera?

Before we begin, here is a list of things you will need:

- 1) 1 Canon 5D MarkIII running official Canon firmware version 1.1.3
- 2) Magic Lantern firmware files, which you can download from <http://dgit.in/18g20a8>. Let's call this "File-Set-A"
- 3) The Dual ISO module which you can grab from <http://dgit.in/1czfgvz>. Download the whole thing as a zip file. Let's call this "File-Set-B."
- 4) An SD card, 8GB or less, with a minimum write-speed of 30MB/s

Getting the Firmware onto the camera

- 1) Once you have downloaded the ML firmware, extract it to a folder on your desktop into a folder called "Custom Firmware." You can call the folder anything you like, but we like to call ours "custom."
- 2) In the folder, you will see three files, one of which is a PDF file for users (this



Explore the firmware menu to discover all the new features of Magic Lantern

doesn't need to go on the camera). The other two are "ml-5D3-113.fir" (this is an actual custom firmware file) and a folder called ML (with several subfolders). Within the ML Folder, create a new folder called "MODULES" (make sure it's all caps).

- 3) Now open Fil-Set-B.zip and delete the 7D folder (you don't need this). Now copy all the files with the extension ".MO" into the Modules folder you created earlier. Copy autoexec.bin and place it in the root (outside the ML folder). It should be giving company to the firmware file.
- 4) Now you have your package ready, so plug in your SD card into your PC. Use a card reader (plugging your camera just won't work)
- 5) Copy the firmware file, autoexec.bin and the ML folder to the ROOT of your SD card.
- 6) Eject SD card and insert into camera.
- 7) Turn on camera and go to the update firmware option. This you can do by pressing "Menu> Settings (the yellow wrench icon)> Fourth page > Update firmware.
- 8) Hit update firmware
- 9) Pray to god your 5D MarkIII doesn't explode from supercharging the living shiznits out of it. (No seriously!)
- 10) Wait for the red lights to stop blinking. Once they stop blinking, wait for a few minutes just to be sure. Now your firmware is installed.

Using the Firmware

- 1) Magic Lantern usage instructions say that once the firmware is installed, pressing the "Delete" button on your camera would bring up the menu, but that instruction is for sanctioned firmware and DOES NOT APPLY to this particular, highly experimental one.
- 2) If you press the Delete key right now, nothing will happen. Turn off the camera at this point and turn it back on.
- 3) Press the Delete key
- 4) Go crazy as the Magic Lantern welcome screen welcomes you.
- 5) Go to "Modules" tab (the last one on the right) and enable "Autoload Modules on Startup." All dots should turn green.
- 6) You can enable Dual ISO from the Movie or Shoot menus and set the corresponding values for the two ISOs you want the camera to shoot at.

You now have successfully loaded the custom firmware onto your 5D Mark III. We have been using it for over a week, with the camera shooting almost everything on Dual ISO mode and the results are actually quite nice. However, given the highly experimental nature of the firmware, we strongly recommend waiting till a more stable version of the firmware is made available. But if you absolutely must use this, we have a guide to using the Dual ISO mode up on our site (along with sample photos and videos) here: <http://dgit.in/15jdtSj>